

# TrES-3b

R-2711

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_190604 · created 2026-09-01T01:10:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458638.7024 +/- 0.0012 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.194 +/- 0.019                 |
| Transit depth (Rp/Rs)^2                           | 3.76 +/- 0.73 [%]               |
| Orbital Inclination (inc)                         | 81.61 +/- 0.6                   |
| Airmass coefficient 1 (a1)                        | 1.187 +/- 0.014                 |
| Airmass coefficient 2 (a2)                        | -0.1379 +/- 0.0069              |
| Scatter in the residuals of the lightcurve fit is | 1.19 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 4.15                            |
| Optimal Annulus                                   | 7.56                            |
| Transit Duration (day)                            | 0.0561 +/- 0.0061               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.194 ± 0.019 vs 0.16309 (deviation 1.62σ) |
| Duration fit vs published | 0.0561 d vs 0.0567 d (deviation 0.09σ)     |
| Mid-time O–C              | 2.4 min                                    |
| Depth SNR                 | 10.1                                       |
| Residual scatter          | 1.19 %                                     |

## Light curve

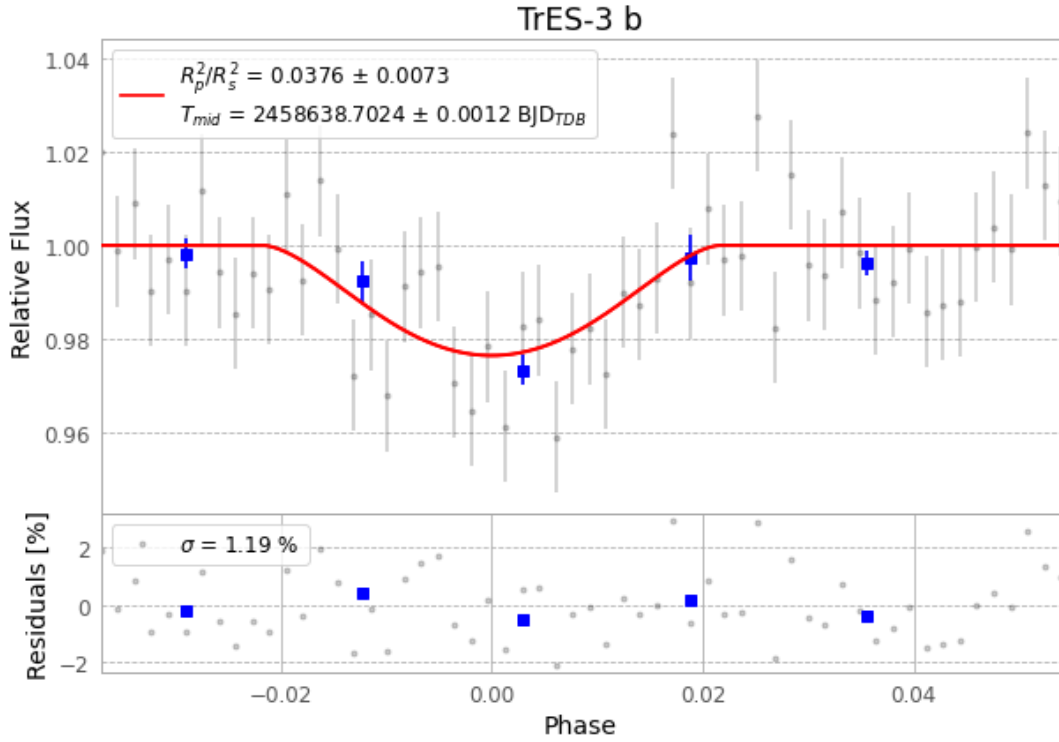


Figure 1 — FinalLightCurve\_TrES-3 b\_2019-06-03.png (SHA-256 107604cf7adfc450...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 82], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 b8b92881885a45b8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

58 FITS frames (38 MB), TRES-3190604033612.FITS → TRES-3190604062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190604.log (9985b8900e62...), FinalLightCurve\_TrES-3 b\_2019-06-03.png (107604cf7adf...), FinalLightCurve\_TrES-3 b\_2019-06-03.csv (b57b2f3b2de9...)

## Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:10Z.